The University of Texas at Austin
UT High School

090124

Third Grade Science Study Guide
Credit by Exam for Credit Recovery or Acceleration

The exam you are interested in taking is designed to test your proficiency in the relevant subject matter. You should be thoroughly familiar with the subject matter before you attempt to take the exam. This CBE CR/A Study Guide can help you prepare for the exam by giving you an idea of what you need to review. You can check your familiarity level by reviewing the Texas Essential Knowledge and Skills (TEKS) for this course. (See below.) To refine your skills, you can refer to any of the state-adopted textbooks.

Texas Essential Knowledge and Skills (TEKS)

Every question that appears on this exam is derived from the knowledge and skills statements and student expectations within the Texas-mandated standards, the Texas Essential Knowledge and Skills (TEKS). You can view the TEKS for this exam online via the following link: https://texreg.sos.state.tx.us/public/readtac$ext.TacPage?sl=R&app=9&p_dir=&p_rloc=&p_tloc=&p_ploc=&pg=1&p_tac=&ti=19&pt=2&ch=112&rl=5.
Refer to section (b), Knowledge and skills, 1A–13B. Throughout this guide, you'll see TEKS references. These refer to the numbers listed under (b) Knowledge and skills; for example, 1A or 3B.

Materials Needed

You will need to bring a #2 pencil to complete the exam. You will receive a computer-graded answer sheet when you arrive at the testing center.

Exam Structure

You will be allowed **3 hours** to complete this exam. The 4th Science exam consists of 40 multiple-choice questions worth a total of 40 points. The exam covers a wide variety of topics. To help you study, we have isolated 5 key topics and provided study tips and sample questions for each.

Topic 1: Scientific and Engineering Practices and Recurring Themes
Topic 2: Matter and Energy
Topic 3: Force, Motion, and Energy
Topic 4: Earth and Space
Topic 5: Organisms and Environment

Scholastic Honesty

When you arrive at the testing center you will be asked to carefully read the exam rules and sign a statement agreeing to take the exam in accordance with the rules. This is called the Examinee's Certification. The following is a copy of these rules:

Examinee's Certification

This certification must be signed *before* the exam is administered and then returned with the completed examination attached, or credit for the exam will not be given.

Scholastic dishonesty is a serious academic violation that will not be tolerated. Scholastic dishonesty encompasses, but is not limited to:

- copying from another student's work;
- using an unauthorized testing proctor or taking the exam at an unauthorized testing location;
- using materials not authorized by a testing proctor;
- possessing materials that are not authorized by a testing proctor, such as lessons, books, or notes;
- knowingly using or soliciting, in whole or Topic, the contents of an unadministered test;
- collaborating with or seeking aid from another student without authorization during the test;
- substituting for another person, or permitting another person to substitute for oneself, in taking a course test or completing any course-related assignment;
- using, buying, stealing, or transporting some or all of the contents of an unadministered test, test rubric, homework answer, or computer program.

Evidence of scholastic dishonesty will result in a grade of *F* on the examination and an *F* in the course (if applicable).

At the testing center, you will be asked to sign a statement that says you have read the above and agree to complete the examination with scholastic honesty.

General Study Tips

The bulleted lists and sample questions in this study guide can assist you in preparing for the exam. It is a fairly complete guide for studying, but does not cover every item on the test. Ultimately, you should use the TEKS to guide your exam preparation.

Additional Study Tips

The following information provides direction for your studies. For each part, you will find study tips and sample questions to give you a general idea of the types of questions you can expect to see on the exam.

Topic 1: Scientific and Engineering Practices and Recurring Themes

This topic relates to your knowledge of safety and the scientific method as it applies to lab and field investigations. It also addresses the use of critical thinking, scientific reasoning, and problem solving to make informed decisions.

Study Tips for Topic 1:

This topic relates to TEKS 1A–5G. Familiarize yourself with those TEKS, and then be prepared to demonstrate knowledge of the following topics:

- demonstrate safe practices during lab and field investigations;
- make informed choices in the use and conservation of natural resources by recycling or reusing materials such as paper, aluminum cans, and plastics;
- plan and implement descriptive investigations, including asking and answering questions, making inferences, and selecting and using equipment or technology needed, to solve a specific problem in the natural world;
- collect data by observing and measuring using the metric system and recognize differences between observed and measured data;
- construct maps, graphic organizers, simple tables, charts, and bar graphs using tools and current technology to organize, examine, and evaluate measured data;
- analyze and interpret patterns in data to construct reasonable explanations based on evidence from investigations;
- demonstrate that repeated investigations may increase the reliability of results;
- communicate valid conclusions supported by data in writing, by drawing pictures, and through verbal discussion;
- in all fields of science, analyze, evaluate, and critique scientific explanations by using empirical evidence, logical reasoning, and experimental and observational testing, including examining all sides of scientific evidence of those scientific explanations, so as to encourage critical thinking by the student;
- draw inferences and evaluate accuracy of product claims found in advertisements and labels such as for toys and food;
- represent the natural world using models such as volcanoes or Sun, Earth, and Moon system and identify their limitations, including size, properties, and materials;
- connect grade-level appropriate science concepts with the history of science, science careers, and contributions of scientists;

- collect, record, and analyze information using tools, including microscopes, cameras, computers, hand lenses, metric rulers, Celsius thermometers, wind vanes, rain gauges, pan balances, graduated cylinders, beakers, spring scales, hot plates, meter sticks, compasses, magnets, collecting nets, notebooks, sound recorders, and Sun, Earth, and Moon system models; timing devices, including clocks and stopwatches; and materials to support observation of habitats of organisms such as terrariums and aquariums;
- use safety equipment as appropriate, including safety goggles and gloves.

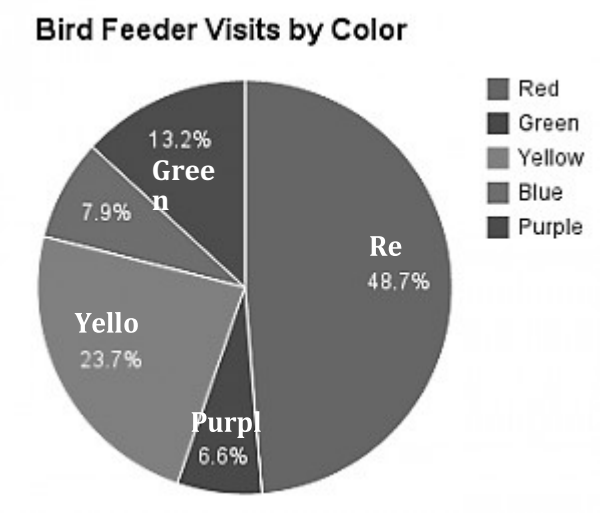
Sample Questions for Topic 1:
The following are sample questions. The correct answers are at the end of this study guide, but try answering the questions without looking at the answers first to check your comprehension.

DIRECTIONS: Select the BEST responses to the following questions.

1. Students create a model of a volcano using clay. They then simulate a volcanic eruption by combining baking soda, vinegar, and red food coloring inside of the clay model. What is a limitation of this model?

    A. The clay model is much smaller than a real volcano.
    B. Clay represents the materials that most volcanoes are made of.
    C. The speed of the reaction in the model is much slower than a real eruption.
    D. Real volcanoes are much more predictable than the baking soda and vinegar reaction.

2. Sheila is an avid birdwatcher, and she wants more birds to visit her feeder. She sets up an experiment to determine if the color of the feeder has an impact over how many birds visit the feeder. Her results of her experiment are shown in the graph below. What valid conclusion can Sheila make using her data?



A. Birds appear to have no preference of color.
B. If Sheila wants less birds to visit her feeder, she should use a red bird feeder.
C. Sheila should use a green or blue bird feeder because they are visited about the same.
D. In order to have the greatest number of birds visit her feeder, Sheila should use a red bird feeder.

Topic 2: Matter and Energy

This topic relates to matter and its measureable physical properties, and how those properties determine how matter is classified, changed, and used.

Study Tips for Topic 2:

This topic relates to TEKS 6A – 6D. Familiarize yourself with those TEKS, and then be prepared to demonstrate knowledge of the following topics:

- measure, test, and record physical properties of matter, including temperature, mass, magnetism, and the ability to sink or float;
- describe and classify samples of matter as solids, liquids, and gases and demonstrate that solids have a definite shape and that liquids and gases take the shape of their container;
- predict, observe, and record changes in the state of matter caused by heating or cooling;
- explore and recognize that a mixture is created when two materials are combined such as gravel and sand and metal and plastic paper clips.

Sample Questions for Topic 2:

The following are sample questions. The correct answers are at the end of this study guide, but try answering the questions without looking at the answers first to check your comprehension.

DIRECTIONS: Select the BEST responses to the following questions.

3. Darren is heating water on a stove. As the water gets hotter and nears its boiling point, what can Darren expect to happen?

   A. The liquid water will condense and turn into gaseous water vapor.
   B. The liquid water will evaporate and turn into gaseous water vapor.
   C. The gaseous water vapor will sublimate and turn into liquid water.
   D. The gaseous water vapor will condense and turn into solid water ice.

4. Madeline is trying to determine the physical properties of several unknown objects. How could Madeline test the objects' magnetism and their ability to sink or float?

   A. She can use a thermometer to see if the objects are magnetic, and she can place them on a pan balance to see if they sink or float.
   B. She can observe the objects with a hand lens to see if they are magnetic, and she can test them with a computer to see if they sink or float.
   C. She can touch the objects with a magnet to see if they are magnetic, and she can place them in a container of water to see if they sink or float.
   D. She can put the objects on a triple beam balance to test their magnetism, and she can place them in a container of vinegar to see if they sink or float.

Topic 3: Force, Motion, and Energy

This topic relates to the many forms of energy, and how they can be observed in cycles, patterns, and systems.

Study Tips for Topic 3:

This topic relates to TEKS 7A – 8B. Familiarize yourself with those TEKS, and then be prepared to demonstrate knowledge of the following topics:

- explore different forms of energy, including mechanical, light, sound, and heat/thermal in everyday life;
- demonstrate and observe how position and motion can be changed by pushing and pulling objects to show work being done such as swings, balls, pulleys, and wagons;
- observe forces such as magnetism and gravity acting on objects.

Sample Questions for Topic 3:

The following are sample questions. The correct answers are at the end of this study guide, but try answering the questions without looking at the answers first to check your comprehension.

DIRECTIONS: Select the BEST responses to the following questions.

5. Troy is exploring the different forms of energy involved in playing an acoustic guitar. What are two forms of energy that he will observe when he plucks the guitar strings?

   A. thermal and sound energy
   B. electrical and light energy
   C. mechanical and sound energy
   D. mechanical and light energy

6. Kyle is playing a carnival game. The goal is to knock down all 10 cans using a small rubber ball. On the first toss, Kyle throws the ball lightly. For his second toss, Kyle pulls his arm back further and throws the ball with more force. Both tosses hit the cans. Which toss probably knocked over more cans and why?

   A. The first toss, because less force will knock over more cans.
   B. The first toss, because more force makes the cans more stable.
   C. The second toss, because more force will knock over more cans.
   D. The second toss, because he pulling his arm back farther gave Kyle better aim.

Topic 4: Earth and Space

This topic relates to the constantly changing surface of the Earth, and that the Earth's surface consists of useful resources.

Study Tips for Topic 4:

This topic relates to TEKS 9A – 11C. Familiarize yourself with those TEKS, and then be prepared to demonstrate knowledge of the following topics:

- explore and record how soils are formed by weathering of rock and the decomposition of plant and animal remains;
- investigate rapid changes in Earth's surface such as volcanic eruptions, earthquakes, and landslides;
- identify and compare different landforms, including mountains, hills, valleys, and plains;
- explore the characteristics of natural resources that make them useful in products and materials such as clothing and furniture and how resources may be conserved;
- observe, measure, record, and compare day-to-day weather changes in different locations at the same time that include air temperature, wind direction, and precipitation;
- describe and illustrate the Sun as a star composed of gases that provides light and heat energy for the water cycle;
- construct models that demonstrate the relationship of the Sun, Earth, and Moon, including orbits and positions;
- identify the planets in Earth's solar system and their position in relation to the Sun.

Sample Questions for Topic 4:

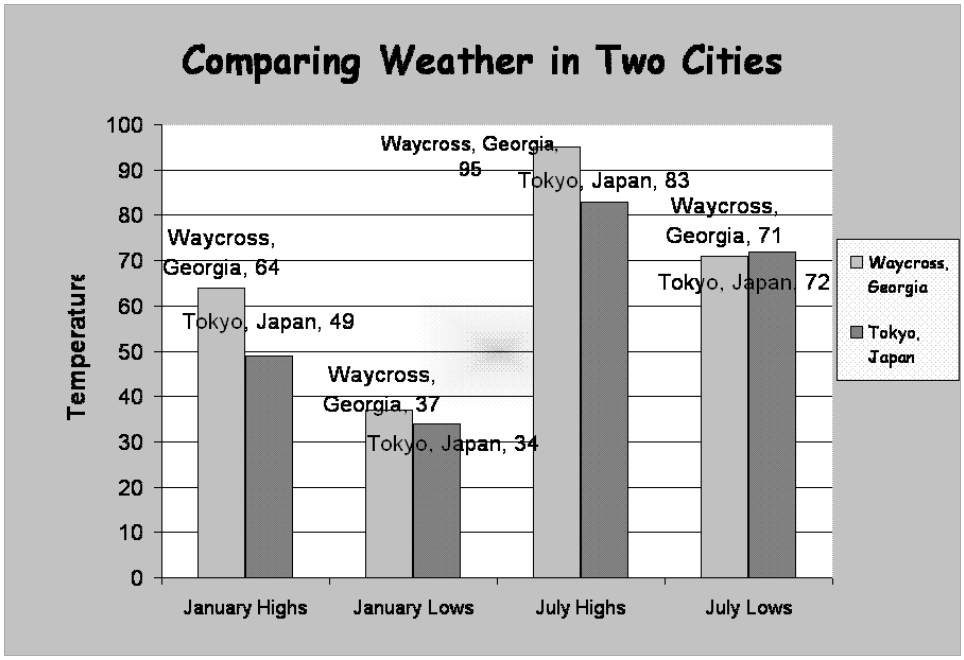
The following are sample questions. The correct answers are at the end of this study guide, but try answering the questions without looking at the answers first to check your comprehension.

DIRECTIONS: Select the BEST responses to the following questions.

7. Which of the following statements about the planets in our solar system is true?

   A. Pluto is farthest from the Sun and Venus is closest.
   B. Mars is farthest from the Sun and Jupiter is closest.
   C. Pluto is closest to the Sun and Neptune is farthest away.
   D. Mercury is closest to the Sun and Neptune is farthest away.

8. How can you compare the weather in the two cities shown in the graph below?



A. Tokyo, Japan is slightly cooler than Waycross, Georgia.
B. Tokyo, Japan is slightly warmer than Waycross, Georgia.
C. It only snows is Tokyo, Japan and not in Waycross, Georgia.
D. The data in this graph does not allow you to compare the weather in these cities.

Topic 5: Organisms and Environment

This topic relates to the relationships, systems, and cycles within environments. In addition it relates to the similar life processes that organisms undergo, and the structures that help them survive within their environments.

Study Tips for Topic 5:

This topic relates to TEKS 12A – 13B. Familiarize yourself with those TEKS, and then be prepared to demonstrate knowledge of the following topics:

- observe and describe the physical characteristics of environments and how they support populations and communities within an ecosystem;
- identify and describe the flow of energy in a food chain and predict how changes in a food chain affect the ecosystem such as removal of frogs from a pond or bees from a field;
- describe environmental changes such as floods and droughts where some organisms thrive and others perish or move to new locations;
- explore how structures and functions of plants and animals allow them to survive in a particular environment;
- explore that some characteristics of organisms are inherited such as the number of limbs on an animal or flower color and recognize that some behaviors are learned in response to living in a certain environment such as animals using tools to get food;
- investigate and compare how animals and plants undergo a series of orderly changes in their diverse life cycles such as tomato plants, frogs, and lady bugs.

Sample Questions for Topic 5:

The following are sample questions. The correct answers are at the end of this study guide, but try answering the questions without looking at the answers first to check your comprehension.

DIRECTIONS: Select the BEST responses to the following questions.

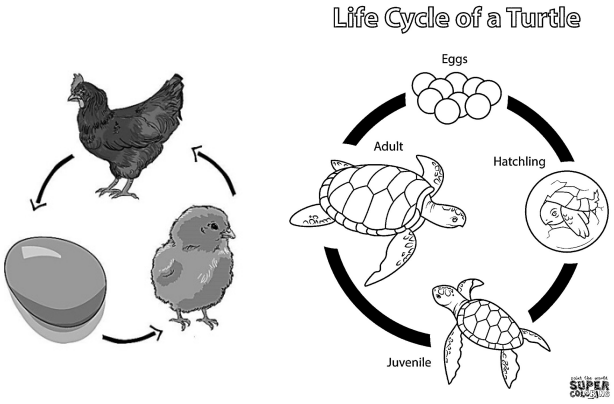
9. During a flood, fire ants will form a floating island to keep from drowning. When the flood is over and the waters recede, what will have happened to the fire ants?

   A. The ants will have moved to a new location.
   B. The ants will have created a bridge to span the flood waters.
   C. The ants will take on a new role of producer in their food web.
   D. The ants will perish because they are no longer in the same ecosystem.

10. How do brightly colored bodies and poisonous skin allow a poison dart frog to survive in its environment?

    A. The frog injects colored poison into its predators in order to keep themselves alive.
    B. The frog's bright coloring warns predators that its skin is poisonous and can kill them.
    C. The frog is endangered because its bright colored body and smelly skin attracts predators.
    D. The frog uses its bright colors to blend into its environment. It kills its prey using its poisonous tongue.

11. How is the life cycle of a chicken similar to the life cycle of a sea turtle?



    A. The turtle life cycle includes four stages, and the chicken life cycle has only three.
    B. Both the chicken and the sea turtle lays eggs that metamorphose from larva to pupa.
    C. Both the chicken and the sea turtle lay eggs that hatch after a specific period of time.
    D. There are no similarities between the life cycle of a chicken and the life cycle of a turtle.

Answer Key

Item Number	Correct Answer	TEKS expectation
1	A	4A
2	D	1F
3	B	6C
4	C	6A
5	C	8A
6	C	7B
7	D	9B
8	A	9A
9	A	12C
10	B	13B
11	C	13B